

Solution

SOUND-02

Class 08 - Science

Section A

1. Read the text carefully and answer the questions:

The to and fro motion of an object is known as vibration. This motion is also called oscillatory motion.

Take a metallic tumbler and a tablespoon. Strike the tablespoon gently at the brim of the tumbler. Hear the sound produced. Now bang the spoon on the tumbler and hear the sound produced again. Is the sound louder when the tumbler is struck hard.



*Thermocol ball
touching the vibrating glass tumbler*

- (i) **(d)** amplitude

Explanation: amplitude

- (ii) **(b)** 60 vibrations per minute

Explanation: 60 vibrations per minute

- (iii) **(b)** hertz

Explanation: hertz

- (iv) 1. frequency

- (v) **(a)** True

Explanation: True

2. Read the text carefully and answer the questions:

Sunita has buy a new house near industrial area. She is very fed-up as during day as well as night unnecessary sound come. They are the loud sound which usually irritate her. She want to create awareness so she gathered the neighbouring people and write slogan about what are the harmful effect of the sound and she also filles complaint about this.

- (i) **(a)** Reading book

Explanation: Reading book

- (ii) **(a)** Option (c)

Explanation: Both Noise producing industries should be set up away residential area and Use of automobile horns should be minimised.

- (iii) **(d)** All of these

Explanation: All of these

- (iv) 1. Noise pollution

- (v) **(a)** True

Explanation: True

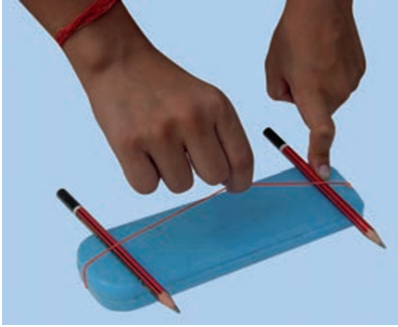
3. Read the text carefully and answer the questions:

An activity is been demonstrated by the teacher in the class in which she follows the following step

Step 1 - Take a rubber band. Put it around the longer side of a pencil box

Step 2 - Insert two pencils between the box and the stretched rubber.

Step 3 - Now, pluck the rubber band somewhere in the middle.



- (i) **(b)** Sound is Produced by a vibrating Body
Explanation: Sound is Produced by a vibrating Body
- (ii) **(b)** All of these
Explanation: All of these
- (iii) **(b)** Energy
Explanation: Energy
- (iv) 1. vibration
- (v) **(a)** True
Explanation: True

4. Read the text carefully and answer the questions:

Rena wants to understand how sound are produced through vocal cord so she performed following activity

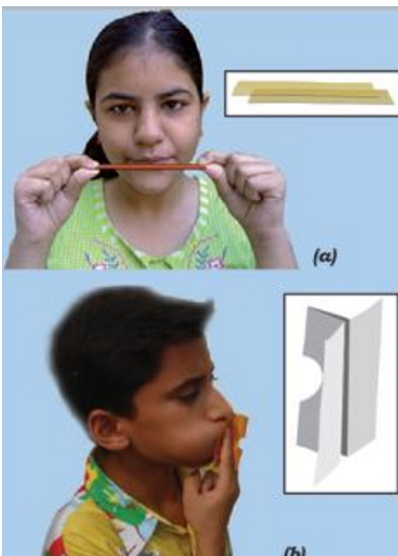
Step 1 - Take two rubber strips of the same size.

Step 2 - Place these two pieces one above the other and stretch them tight.

Step 3 - Now blow air through the gap between them. As the air blows through the stretched rubber strips, a sound is produced.

Step 4 - You can also take a piece of paper with a narrow slit and hold it between your fingers.

Step 5 - . Now blow through the slit and listen to the sound.



- (i) **(b)** Larynx
Explanation: Larynx
- (ii) **(b)** hearing organ
Explanation: hearing organ
- (iii) **(a)** 20 Hz to 20,000 Hz
Explanation: 20 Hz to 20,000 Hz
- (iv) 1. Vocal cords
- (v) **(a)** True
Explanation: True

5. Read the text carefully and answer the questions:

Anita take a bucket or a bathtub. She fills it with clean water. Take a small bell in one hand. She then shakes this bell inside the water to produce sound. Make sure that the bell does not touch the body of the bucket or the tub.



- (i) (c) vacuum

Explanation: vacuum

- (ii) (b) Amplitude

Explanation: Amplitude

- (iii) (a) Sound needs medium propagation to travel

Explanation: Sound needs medium propagation to travel

- (iv) 1. strings

- (v) (b) False

Explanation: False

Section B

6. Column A	Column B
(i) Black and forth motion	(b) Vibration
(ii) Ektara	(D) Musical instrument
(iii) Voice box	(e) Larynx
(iv) Eardrum	(a) Thin membrane
(v) Loudness	(c) Decibel

7. a. - iii
b. - iv
c. - ii
d. - i

8. a. - iii
b. - iv
c. - ii
d. - i

9. a. - iii
b. - iv
c. - ii
d. - i

10. a. - iii
b. - iv
c. - ii
d. - i

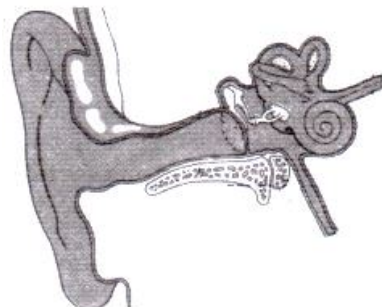
11. a. - iii
b. - iv
c. - ii
d. - i

12. Column A	Column B
(i) Unit of frequency	(d) Hertz
(ii) Low frequency	(e) Low shrillness

(iii) Sound	(a) Vibrations of the body
(iv) Pleasant sounds	(b) Music
(v) Unwanted sounds	(c) Noise

Section C

13. The hearing organs in humans are. Ears have a stretched structure called eardrum. When ears receive sound, it enters down the ear canal and reach eardrum. Vibrations of sound vibrate the eardrum and it sends vibrations to inner ear from which vibrations are sent to brain for interpretation of sound.



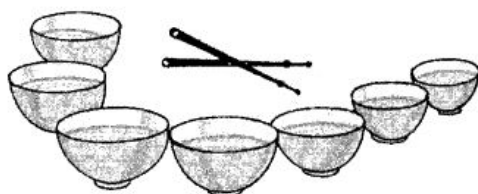
Human ear.

14. Take a metal glass tumbler. Place a cellphone in it. Ask your friend to give ring on 1 cellphone. Listen to the ring carefully. Now surround the rim of the tumbler with your hands. Put your mouth on opening between your hands. Indicate to your friend to give a ring again. Listen the ring while sucking air from the tumbler. The sound become very less. This activity shows that sound requires a medium to propagate.



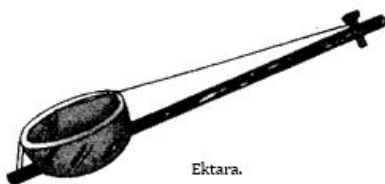
Sound needs a medium to travel.

15. Since the range of frequency for every vibrating body is different. But we can hear the vibrations which lie between the range of frequencies from 20 Hz to 20000 Hz. So, the sound of every vibrating body cannot be heard by us.
16. Jaltrang works on the principle that the quality of sound is changed as its frequency changes. Metal tumbler filled with water up to different height, give vibrations of different frequencies, hence produce different types of sounds.



Jaltrang.

17. It is better to buy the house which is three lanes away from the roadside as there will be less noise as compared to the one on the main road. The noises can be caused by vehicles. As the distance between the source and the listener increases, the intensity of noise decreases. So it will be advisable to buy the house which is three lanes away.
18. Take a hollow coconut shell or an earthen pot and make a musical instrument. Such instrument is called ektara. Play this instrument, you hear a sound. The stretched rubber band or a wire is its vibrating part.



Ektara.

19. Harmful Effects Of Noise Pollution

- Noise pollution can cause irritation and loss of concentration.
- Noise pollution can cause anxiety, sleep disturbance, stress and high blood pressure.
- Noise pollution can also cause ear damage and loss of hearing if one has continuous exposure to noise over a long period of time.

20. Presence of unwanted and excessive sounds in the environment is called noise pollution. Noise is produced by the irregular vibrations of the sound producing source. It causes discomfort to us. Excessive sounds are produced by honking of horns, loudspeakers, crackers and machines.
21. (a) Stretched membrane
(b) String of sitar
(c) Air column.
22. The vibrating objects produce sound. In some cases the vibrations are easily visible to us. But in most of the cases, the vibrations are not visible due to very small amplitude. But the vibrations in all the cases can be felt.
23. Number of vibrations per second = 500
Time period = time taken for one vibration = $\frac{\text{Total time}}{\text{Number of vibration}} = \frac{1}{500} \text{ Hz} = 0.002 \text{ Hz}$
24. It will create more loud sound in case of string stretched with a greater force because amplitude is greater in this case and loudness depends on the amplitude. Greater will be the loudness, greater will be the amplitude.
25. **Measures to control noise pollution**
- 1) We should not play radio, stereo systems and televisions too loudly.
 - 2) The horns of motor vehicles should not be gone unnecessarily.
 - 3) The bursting of crackers should be avoided.
 - 4) The noise making factory and airports should be shifted away from the residential area of the city.
 - 5) Loud speakers should be played at low volume during marriages and other social function.
 - 6) Trees should be planted along the roads and around building to reduce the noise pollution from the roads.
26. Following are the major sources of noise pollution :
- (i) Sounds of vehicles.
 - (ii) Sounds of bursting of crackers and explosives.
 - (iii) Sounds of loudspeaker, TVs, transistors.
 - (iv) Sounds of kitchen appliances.
 - (v) Sounds of desert coolers, air conditioners etc.
27. We know that
Distance = Speed $\times$ Time
Distance = $330 \times 5 = 1650 \text{ m} = 1.65 \text{ km}$ ($\because 1 \text{ km} = 1000 \text{ m}$)
So, she is standing at 1.65 km from the place where lightning occurs.
28. Sources of noise pollution are as follows:
- i. Horns of vehicles
 - ii. Loudspeakers of temples
 - iii. Generators running without silencers.
29. Light travels at the speed of $3 \times 10^8 \text{ m/s}$ while sound travels at a speed of 332 m/s. The speed of light much more than the speed of sound. Due to this, light reaches to us faster than sound. Hence, during lightning we see the streak of light earlier than hearing the sound of thunder.
30. Sound is produced by voice box or larynx; in human beings. The voice box or larynx is situated in the upper part of wind pipe. There are two vocal cords stretched across the larynx in way that there is a small gap between them. When air is forced through the gap, vocal cords begin to vibrate and sound is produced.
31. Echolocation is nothing but just a reflection of sound. For example- when you throw a ball on a wall, it hits the wall and bounces back. Same is the case with sound waves. When sound wave travels, it hits things which come in its path and bounces back. This phenomenon is called reflection of sound or echo. To hear the echo, one should be at 17m distance from the reflected surface.
32. Noise: The unpleasant sounds are called noise. It is unwanted sound. It is our ears.
Musical sound: The sound which produces pleasant sensations is called musical sound. It is produced by various musical instruments.
33. i. Noise pollution damages the ears. Constant loud music can harm hearing or causes permanent damage to ears.
ii. It becomes the reason of poor health like lack of sleep, hypertension, frustration etc.
iii. Noise pollution has effects on the environment
34. Sound which is pleasant to ear is called Music. We enjoy the sounds from musical instrument and this is known as a musical sound. They are produced by the regular vibrations of the sound producing source. All the musical instruments produce musical sounds. The musical instruments have strings or the stretched membranes attached or fixed to them. Some instruments have air columns. When these instruments vibrate, the sound is produced.

35. Place a squeaking toy or sound making toy in a polythene bag and immerse it in a bucket full of water. If we put our ear to the side of this bucket and press the toy, we can hear the sound of squeaking toys clearly. The sound of squeaking toy comes to ear through water contained in the bucket. This shows that sound can travel through water, which is a **liquid**.



Sound travelling through water

36. Music are sounds which are pleasant to hear. Sounds from flutes, pianos and violins are pleasant to hear.

Noise are sounds which are unpleasant to hear.

Sounds which are unpleasant to hear are:

- i. Sounds from bus horns and truck horns.
- ii. Electrical generator sounds.
- iii. Gun shot sounds.
- iv. Jackhammer sounds

Yes, sometimes when the music are played at high volumes, it becomes noise.

37. The number of oscillations per second is called frequency. Its unit hertz. A frequency of 1 Hz is one oscillation per second. The frequency determines the shrillness of sound. It is also called pitch of sound. The high frequency makes the pitch high and if the pitch is low, the frequency is less.

38. Given, number of oscillations = 10

Time taken = 20s

As, we know that the number of oscillations per second is frequency.

$$\therefore \text{Frequency} = \frac{\text{Number of oscillations}}{\text{Time taken}} = \frac{10}{20} = 0.5 \text{ Hz}$$

$$\therefore \text{Time period} = \frac{1}{\text{Frequency}} = \frac{1}{0.5} = 2\text{s}$$

39. **(i) Amplitude:** It is the maximum displacement from the mean position of an oscillatory body. It controls the loudness of a sound; more the amplitude, more loud the sound is.

(ii) Frequency: The Number of oscillations in one second is called frequency. It controls the pitch of the sounds. High pitched sound means high frequency.

40. The vibrating bodies produce the sound. When a body vibrates, sound is produced. If vibrations are stopped the sound is not produced any more.

41. Take a long metal rod or a metre scale and hold its one end to your ear. Ask your friend to gently scratch or tap at the other end of the metal rod. You can hear the sound. The sound reaches your ear through the solid rod. The other friends cannot hear the sound because sound is very low. This activity shows that sound travels in solids also.



Sound travelling through a metre scale.

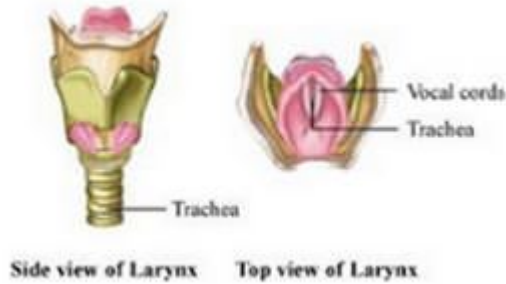
42. The decreasing amount of air causes decrease in the loudness of sound. When air is removed completely from a vessel, it is said that there is a vacuum. If all the air is sucked from the vessel, the sound would stop completely. The sound cannot travel through a vacuum.

43. Loudness of sounds is proportional to the square of amplitude of the vibrations producing the sounds. If the amplitude becomes twice, the loudness increases by a factor of four. The unit of loudness is decibel (dB). The sound above 80 dB becomes physically painful. The loudness of sound depends on the amplitude. When the amplitude of vibration is large the sound produced is loud, when the amplitude is small the sound produced is feeble.

44. No, we cannot hear any sound on the surface of the moon. This is because moon has no air and no atmosphere or the medium whereas sound needs the medium to travel. Sound waves cannot propagate if there is no material medium.

45. In humans, the sound is produced by the voice box or the larynx. It is situated in the throat. Below it is the wind pipe. The two vocal box or larynx in such away that it leaves a narrow slit between them for the passage of air when the lungs force air through

the slit, the vocal cords vibrate producing sound.



46. **Frequency** of wave is defined as number of oscillations completed by the wave in one second. SI unit to measure frequency is Hertz. Frequency also determines the pitch or shrillness of wave. If the frequency of sound wave is high then the sound is shrill and the pitch is high, whereas if the frequency is low then the sound have a lower pitch.
Examples of higher pitch sound are sound produced by whistle and sound produced by bird.
Examples of low pitch sound are roar of lion and men's voice.
47. Yes, the frying pan will vibrate. Since it is being hit by the stick but vibrations need a medium to travel and there is no medium in a vacuum, so we cannot hear the vibrations produced.
48. No, they cannot talk to each other without using any special device because there is no medium in space and sound needs the medium to travel.
49. **Measures to control noise pollution**
- 1) We should not play radio ,stereo systems and televisions too loudly.
 - 2) The horns of motor vehicles should not be gone unnecessarily.
 - 3) The bursting of crackers should be avoided.
 - 4) The noise making factory and airports should be shifted away from the residential area of the city.
 - 5)loud speakers should be played at low volume during marriages and other social function.
 - 6) Trees should be planted along the roads and around building to reduce the noise pollution from the roads.
50. Speed of light travels faster than the speed of sound. Therefore Boojho sees the light way before hearing the sound of the cracker.
51. Take a rubber band, put it around the longer side of a pencil box. Insert two pencils between the box and stretched rubber. Pluck the rubber band somewhere in the middle. You hear a sound and also see that rubber band vibrates. This activity shows that vibrating objects produce sound.

